Problem Solving through Programming (CSE103)
Multiple Choice Questions
Unit- 4

Q.No		
1`	Which operator is used to access the address of a variable in C?	
	a) *****	**b) &**
	c) @	d) %

2	What will be the output of the following code?
	int main()
	{
	int var = 7;
	int *p;
	p = &var;
	printf("%d", *p);
	}	

	a)7	b) The memory address of var
	c)0	d)Undefined

3	What does the following code snippet do?
	int main()
	{
	int arr[5] = {1, 2, 3, 4, 5};
	int *ptr = arr;
	printf("%d", *(ptr + 2));
	}	

	a)Prints 2	**b) Prints 3**
	c) Prints the address of the third element	d) Prints 4
4	What is the correct way to access the value of a variable through a pointer?	
	a) &ptr	**b)*ptr**
	c)ptr*	d)@ptr
5	What is a pointer in C?	
	a) A variable that stores a character.	b) A variable that stores a floating-point number.
	c) A variable that stores the memory address of another variable.	d) A variable that stores a Boolean value.
6	Which format specifier is used to print the address stored in a pointer?	
	a)%d	**b)%p**
	c)%f	d)%u
7	Which of the following correctly declares a pointer to an integer in C?	
	a) int p;	b) **int *p;**
	c) int &p;	d) pointer int p;
8	Which operator is used to get the value stored at a pointer's address?	
	a) *	b) &
	c) @	d) %

9	What will the following print?
	int x = 5;
	int *p = &x;
	printf("%p", p);	

	a) Value of x	**b) Address of x**
	c) Address of pointer p	d) Garbage value

10	What does this print?
	int a = 5, b = 10;
	int *p = &a, *q = &b;
	*q = *p;
	printf("%d %d", a, b);	

	a)5 5	b)5 10
	c)10 5	d)Compilation Error
11	Which of the following is a null pointer in C?	
	a) int *p = NULL;	b) int *p = 0;
	c) Both a and b	d) only A
12	Which of the following pointer declarations is invalid?	
	a) int *ptr1;	b) float *ptr2;
	c) char *ptr3;	**d) int ptr4*;**
13	What happens if you dereference a NULL pointer in C?	
	a) Prints 0	**b) Segmentation fault/runtime error**
	c) Prints NULL	d) Undefined compile-time behavior
14	Which of the following is **NOT** a valid pointer type in C?	
	a) int*	b) void*
	c) char**	**d) string***
15	Which of the following is **not** a valid storage class in C?	
	a) auto	b) static
	c) volatile	**d) public**
16	What is the main advantage of recursion?	
	a) Uses less memory	**b) Makes code shorter and easier to understand for some problems**
	c) Executes faster than loops	d) Avoids stack usage
17	Which keyword is used to define a function in C?	
	a) def	b) func
	c) function	**d) No special keyword is required**
18	Which of the following is a correct function prototype in C?	
	a) function sum(int, int);	b) sum int(int, int);
	c) int sum(int, int);	d) int sum;
19	Which of these is **not** a valid way to call a function in C?	
	a) func();	b) func(x);
	c) x = func();	**d) call func;**
20	Which function is always the first to execute in a C program?	
	a) start()	**b) main()**
	c) init()	d) first()
21`	What is the correct syntax for declaring a user-defined function in C that returns an int and takes two float parameters?	
	a) int func(float, float);	b) func int(float, float);
	c) int func(float x, y);	d) float func(int, int);

22	What will be the output of the following C program?
	
	`#include <stdio.h>`
	`void greet() {`
	`  printf("Hello");`
	`}`
	`int main() {`
	`  greet();`
	`  greet();`
	`  return 0;`
	`}`	

	a) Hello	**b) HelloHello**
	c) Hello Hello	d)Compilation Error
23	Which storage class in C makes a variable retain its value between function calls?	
	a) auto	b) register
	c) static	d) extern

24	What will be the output of this program?
	`#include <stdio.h>`
	`void counter() {`
	`  static int count = 0;`
	`  count++;`
	`  printf("%d ", count);`
	`}`
	`int main() {`
	`  counter();`
	`  counter();`
	`  counter();`
	`  return 0;`
	`}`	

	a)1 1 1	**b)1 2 3**
	c)0 1 2	d)Error
25	Which of the following is true about register variables?	
	a) They are stored in CPU registers for faster access.	b) They can be directly accessed using & operator.
	c) They are slower than auto variables.	d) They are always global.

26	What will the following program output?
	
	`#include <stdio.h>`
	`int fact(int n) {`
	`  if (n == 0) return 1;`
	`  return n * fact(n - 1);`
	`}`
	`int main() {`
	`  printf("%d", fact(4));`
	`  return 0;`
	`}`	

	a)24	b)10
	c)4	d)5
27	Recursive functions must have:	
	a) At least two parameters	**b) A base condition to stop recursion**
	c) A loop for termination	d) No return value

28	What will be the output of the following code?
	`#include <stdio.h>`
	`void display(int n) {`
	`  if (n == 0) return;`
	`  printf("%d ", n);`
	`  display(n - 1);`
	`}`
	`int main() {`
	`  display(3);`
	`  return 0;`
	`}`	

	a)3 2 1	b)1 2 3
	c)0 1 2 3	d)No Output
29	What will happen if a recursive function does not have a base case?	
	a) It will return 0	b) It will stop automatically after 10 calls.
	c) It will result in infinite recursion and possible stack overflow.	d) It will compile but not run.

30	What will the following code print?
	`#include <stdio.h>`
	`int sum(int n) {`
	`  if (n == 1) return 1;`
	`  return n + sum(n - 1);`
	`}`
	`int main() {`
	`  printf("%d", sum(5));`
	`  return 0;`
	`}`	

	a)15	b)5
	c)10	d)6